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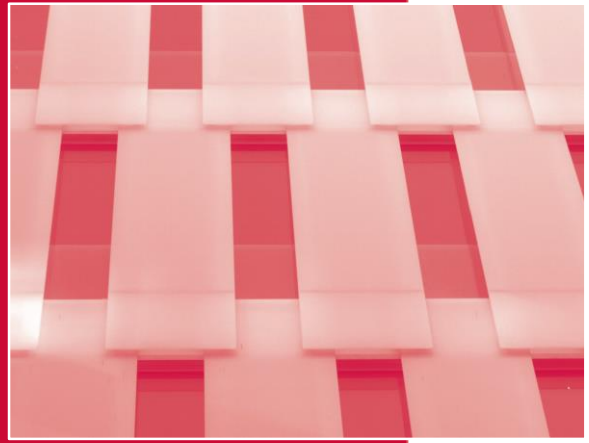
Financial Accounting – Course 30693



Financial Statement presentation, reclassification and analysis

Part 2

Università Bocconi
UNDERGRADUATE SCHOOL

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AGENDA

1. Understanding the Business
2. External decision-makers. F/S presentation
3. Internal decision-makers
4. F/S reclassification. Key indicators
5. Ratio Analysis
   - 5A. Ratios: general overview
   - 5B. Profitability ratios
   - 5C. Efficiency ratios
   - 5D. Liquidity ratios
   - 5E. Solvency ratios
6. FSA in the real-life world

PART 2

5A. Ratios: general overview

Why Ratio Analysis Matters?

Ratio Analysis **is an essential tool for performance evaluation and for crisis prevention**.

For example, the Italian "Business Crisis and Insolvency Code" (Codice della Crisi d'Impresa - D.Lgs. 14/2019) requires the use of ratios for the following purposes:

- Early detection of financial distress
- Continuous monitoring of economic and financial equilibrium
- Preservation of going concern

Firms must implement adequate systems to identify:

- Financial imbalances
- Signals of potential crisis

Ratio analysis enables:

- Monitoring of liquidity, solvency, and profitability
- Detecting early warning signs of crisis

Ratios: General Overview

Ratio analysis is an analytical tool used to measure the proportional relationship between amounts reported in the financial statements.

Ratios must then be interpreted, which requires experience and judgment.

REMEMBER: when computing ratios:

- Balance sheet amounts represent **A SPECIFIC POINT IN TIME.**
- Income statement amounts relate **TO A PERIOD OF TIME.**

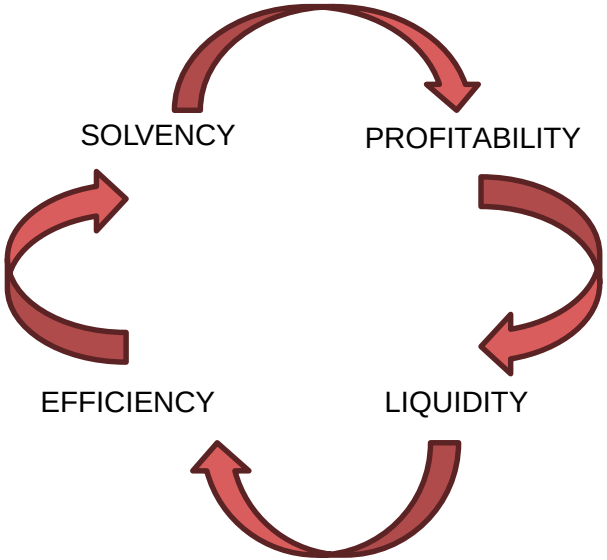
To adjust for this difference, most analysts use the

AVERAGE balance sheet amount

when comparing a balance sheet number to an income statement number.

The main purpose of the Financial Statements analysis is the

ASSESSMENT OF THE PERFORMANCE AND THE SOUNDNESS OF THE FIRM, ACCORDING TO THE FOLLOWING MAIN DIMENSIONS



Some typical questions:

- Is the profit that is being earned satisfactory?
- Is the company able to satisfy its liabilities?
- Is the long-term capital structure suitable?
- Is efficient use being made of assets and resources?
- Is the company growing?

Among the many ratios that can be computed using a company's financial statements, analysts focus on **THOSE THAT ARE MOST USEFUL IN A GIVEN SITUATION.**

The clearer the questions that analysts seek to answer (based on the sector and the research objective), the easier it is to select the most appropriate ratios to compute.

Ratios can generally be grouped into the following categories:

I. Profitability ratios

II. Efficiency ratios

III. Liquidity ratios

IV. Solvency ratios

5B. Profitability ratios

Profitability Ratios

Profitability is a primary measure of a company's overall success. These ratios focus mainly on net income and how it compares to other items reported in the financial statements.

1.Earnings per Share (EPS)

$$\frac{\text{Net Income}}{\text{Weighted Average Number of Common Shares Outstanding}}$$

- Emphasizes the amount of earnings attributable to a single share of common stock.
- There are two types: basic EPS and diluted EPS.
- Comparisons across companies may be misleading due to differences in the number of shares issued.
- One of the most widely reported ratios.
- The only ratio required by GAAP.
- EPS can be affected by the issuance or repurchase of common shares.

2. Return on Equity (ROE)

$$\frac{\text{Net Income}}{\text{Average Total Stockholders' Equity}}$$

- Measures the income earned on shareholders' investment in the business.
- Reflects the company's ability to generate profits for its shareholders.
- Is influenced by financing decisions and other non-operating factors (e.g., treasury shares).

Profitability Ratios

3. Gross Profit Margin Ratio

$$\frac{\text{Gross Profit}}{\text{Net Sales Revenue}}$$

- Higher margins suggest strong differentiation or cost advantage, but must be assessed relative to industry norms
- All else being equal, a higher gross profit results in higher net income.

4. Return on Sales (ROS)

$$\frac{\text{EBIT}}{\text{Net Sales Revenue}}$$

- Measures a company's ability to generate operating profit from sales.
- A good measure of operating efficiency and expense control.
- All else being equal, a higher ROS indicates better operating performance.

5. Net Profit Margin Ratio

$$\frac{\text{Net Income}}{\text{Net Sales Revenue}}$$

- Measures how much of each sales dollar generated during the period is profit.
- The higher the ratio, the more effective the management of sales and expenses.

*These ratios (#3, #4, #5) should not be analyzed in isolation: **companies should always be compared within the same industry** (e.g., the food industry, the jewelry industry, etc.).*

The Home Depot, Inc. Case

Financial Statements

The Home Depot, Inc. is the world's largest home improvement retailer, operating a vast network of big-box stores across North America. Its business model centers on Interconnected Retail, which seamlessly merges a robust digital platform with physical locations that serve as both showrooms and distribution hubs. The company targets two primary customer segments: DIY homeowners and Professional contractors.

The Home Depot, Inc. Case

Income Statement

THE HOME DEPOT, INC. CONSOLIDATED STATEMENTS OF EARNING			
In millions, except per share data	**Fiscal 2020**	**Fiscal 2019**	**Fiscal 2018**
Net sales	$ 132,110	$ 110,225	$ 108,203
Cost of sales	87,257	72,653	71,043
Gross profit	44,853	37,572	37,160
Operating Expenses:			
Selling, general and administrative	24,447	19,74	19,513
Depreciation and amortization	2,128	1,989	1,87
Impairment loss	-	-	-
Total operating expenses	26,575	21,729	21,630
Operating income			
Interest and investment income	(47)	(73)	(93)
Interest expense	1,347	1,201	1,051
Other	-	-	-
Interest and other, net	1,300	1,128	974
Earnings before provision for income taxes	16,978	14,715	14,556
Provision for income taxes	4,112	3,473	3,435
Net Earnings	$ 12,866	$ 11,242	$ 11,121
Basic weighted average common shares	1,074	1,093	1,137
Basic earnings per share	$ 11.98	$ 10.29	$ 9.78
Diluted weighted average common shares	1,078	1,097	1,143
Diluted earnings per share	$ 11.94	$ 10.25	$ 9.73

The Home Depot, Inc. Case

Balance Sheets (Assets)

THE HOME DEPOT, INC. CONSOLIDATED BALANCE SHEETS		
In millions, except per share data	**End of Fiscal 2020**	**End of Fiscal 2019**
Assets		
Current assets		
Cash and cash equivalents	$ 7,895	$ 2,133
Receivables, net	2,992	2,106
Merchandise investories	16,627	14,531
Other current assents	963	1,04
Total current assets	28,477	19,81
Net property and equipment	24,705	22,77
Operating lease right-of-use assets	5,962	5,595
Goodwill	7,126	2,254
Other assets	4,311	807
Total assets	$ 70,581	$ 51,236

The Home Depot, Inc. Case

Balance Sheets (Liabilities & Stockholders' Equity)

THE HOME DEPOT, INC.

CONSOLIDATED BALANCE SHEETS

In millions, except per share data	**End of Fiscal 2020**	**End of Fiscal 2019**
Liabilities and Stockholders' Equity		
Current liabilities:		
Short-term debt	$ -	$ 974
Accounts payable	11,606	7,787
Accrued salaries and related expense	2,463	1,494
Sales taxes payable	774	605
Deferred revenue	2,823	2,116
Income taxes payable	193	55
Current installments of long-term debt	1,416	1,839
Current operating lease liablities	828	828
Other accrued expenses	3,063	2,677
Totale current liabilities	23,166	18,375
Long-term debt, excluding current installments	35,822	28,670
Long term operating lease liabilities	5,356	5,066
Deferred income taxes	1,131	706
Other long-term liabilities	1,807	1,535
Total liabilities	67,282	54,352
Common stock, par value $0.05; authorized: 10,000 shares; issued: 1,789 shares at end of Fiscal 2020, and 1,786 shares at end of Fiscal 2019; outstanding: 1,077 shares at end of Fiscal 2020 and end of Fiscal 2019	89	89
Paid-in capital	11,540	11,001
Retained earnings	58,134	51,729
Accumulated other comprehensive loss	(671)	(739)
Treasury stock, at cost, 712 shares at end of Fiscal 2020, and 709 shares at end of Fiscal 2019	(65,793)	(65,196)
Total stockholders' equity (deficit)	3,299	(3,116)
Total liabilities and stockholders' equity	$ 70,581	$ 51,236

The Home Depot, Inc. Case (Profitability Ratios)

1. EPS

- **$12.866/1.074**
- **= $11.98**

- Average number of shares outstanding is reported on the bottom of the income statement

2. ROE

- **$12.866/$3.299***
- **= 390.00% (rounded)**

- On average, for every $1.00 equity investors contributed to The Home Depot, the company earned $3.90 in fiscal year 2020.

3. Gross Profit Margin Ratio

- **$44.853 /$132.110**
- **= 33.95%**

- After subtracting the direct costs associated with selling products, The Home Depot had approximately 34 cents of each dollar of sales remaining to cover other expenses.

4. ROS

- **$18.278/$132.110**
- **=13.84%**

- EBIT FY 2020 = $44.853-$24.447-$2.128 = $18.278
- After covering operating expenses, The Home Depot retains approximately 14 cents of operating profit per dollar of sales, reflecting solid cost management and operational efficiency.

5. Net Profit Margin Ratio

- **$12.866/$132.110**
- **=9.74%**

- For fiscal year 2020, each dollar of The Home Depot's sales generated over 9 cents of profit. For each dollar of sales, The Home Depot is subtracting just around 90 cents in expenses.

** Normally, average equity is used as the denominator. However, since it was negative in fiscal years 2019 and 2018, the ending equity balance for fiscal year 2020 is used instead (Home Depot had repurchased treasury stock from the market).*

5C. Efficiency ratios

Efficiency Ratios

Efficiency ratios (also called Asset Turnover ratios) are crucial for measuring a company's ability:

- to generate revenues in relation to its total assets;
- to efficiently manage its payables, inventories, and receivables.

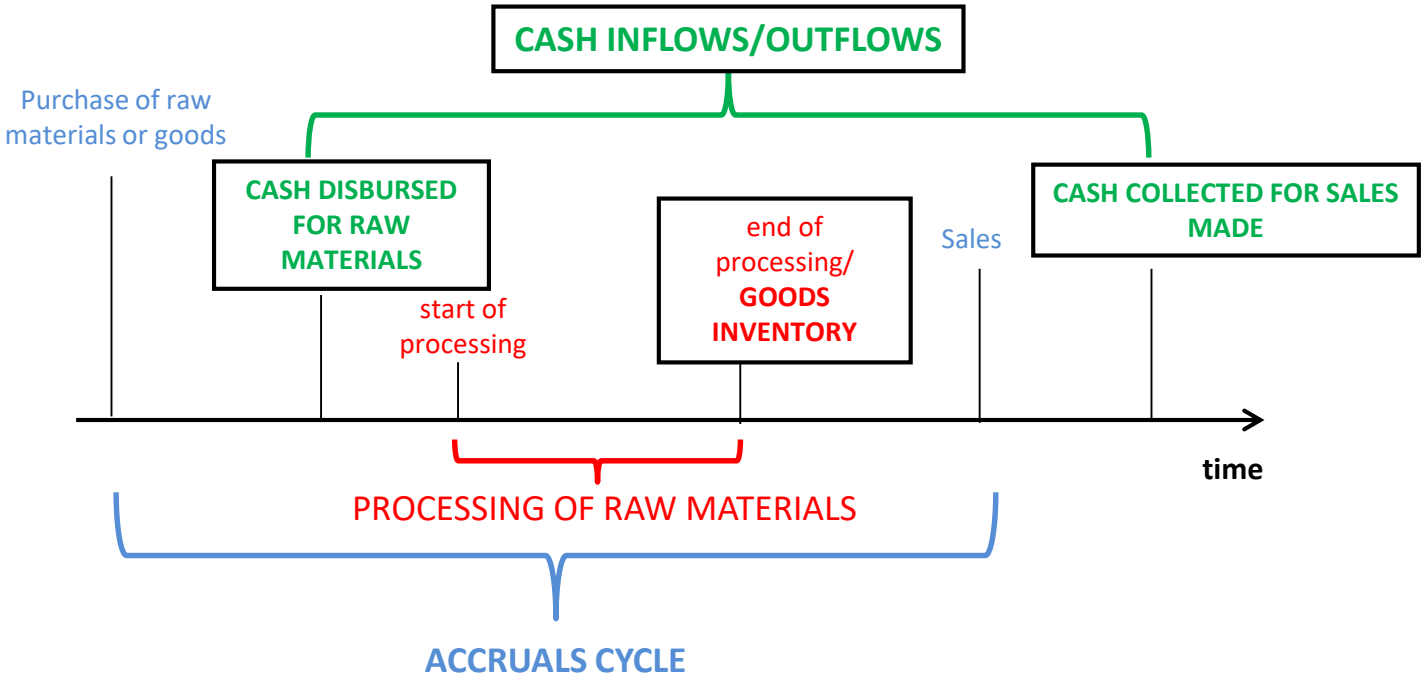
With reference to the latter, the terminology widely used in practice includes DSO, DPO, and DOI: Days Sales Outstanding, Days Payables Outstanding, and Days of Inventory.

Excellence in managing DSO, DPO, and DOI can generate significant benefits for shareholders.

As such, managers and consultants closely monitor day-to-day activities to improve these ratios. This is commonly referred to as working capital management, as inventory, accounts receivable, and accounts payable are the main components of working capital.

Efficiency Ratios

Management of a company's cycles (operating, accruals, and cash) is crucial.



Efficiency Ratios

6.Total Asset Turnover Ratio

$$\frac{\text{Net Sales Revenue}}{\text{Average Total Assets}}$$

- Captures how effectively the company uses its assets to generate revenues.
- The higher the ratio, the more efficient the use of total assets.

7. DSO (also: Average Days to Collect Receivables)

$$\frac{\text{Average Net Receivables}}{\text{Net Credit Sales}} \times 365$$

- When net credit sales are not available, net sales are used as an approximation.
- Indicates the average time (in days) it takes to convert receivables into cash.
- The higher the total, the longer the collection period (thus, a lower value is preferable).
- Impacted by VAT, bad debt allowances, and other contra-revenues (e.g., returns)

Efficiency Ratios

8. DOI – Average Days to Sell Inventory

- $$\frac{\text{Average Inventory}}{\text{Cost of Goods Sold}} \times 365$$

- Indicates the average time (in days) required for the company to sell its inventory.
- The higher the value, the less efficient the production and sales process; therefore, a lower value is preferable.
- Trade-off: availability of goods in inventory vs. efficiency in cash management.
- Impacted by: measurement method (FIFO, LIFO, etc.) and slow-moving inventory (valued at the lower of cost or NRV).

9. DPO – Average Days Payables Are Outstanding

- $$\frac{\text{Average Accounts Payable}}{\text{Cost of Goods Sold}} \times 365$$

- Indicates the average time required to pay suppliers.
- The higher the value, the longer the company takes to settle its accounts payable.
- High value is preferable (the company is able to delay payments to suppliers), but an increasing ratio across years might be signal of low liquidity

Focus on DSO, DPO and DOI

Ceteris paribus, the optimal management of DPO, DSO, and DOI is a lever to increase a company's value.

DSO

- The sooner a company collects cash from its customers, the more cash is available.

DOI

- The less time products remain in inventory, the less time it takes to generate cash from sales.

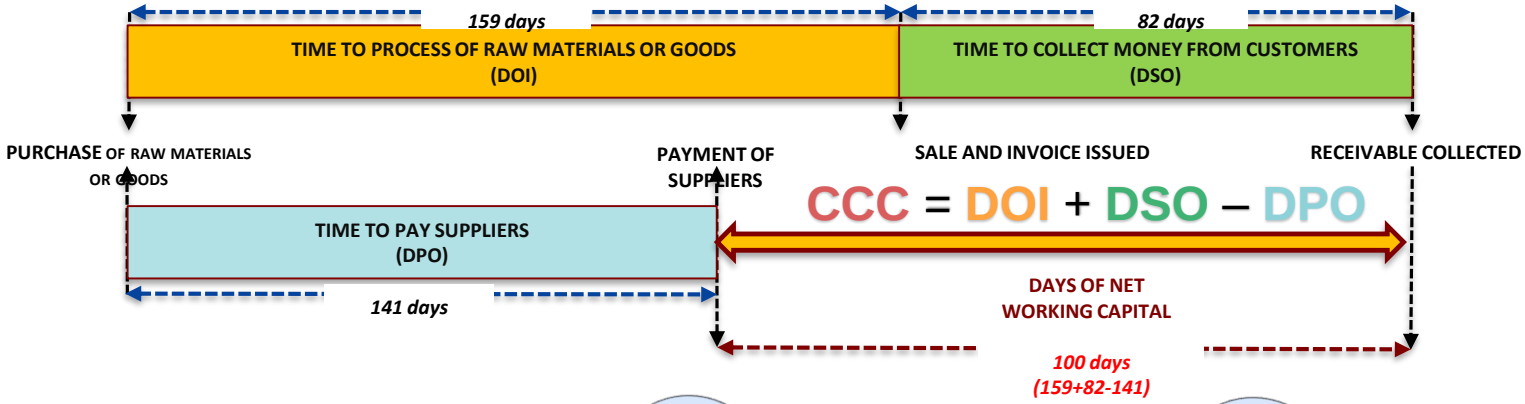
DPO

- The longer it takes to pay suppliers, the better it is.
- If a company has a low DPO, it likely requires external short-term financing (e.g., bank financing), which is interest-bearing (i.e., reduces profit).
- If a company has good DSO and DOI, it can choose to pay suppliers earlier to maintain strong supplier relationships (good suppliers = good operations) and/or request discounts (i.e., higher profit).

The above (especially for DSO and DOI) is valid for any company, regardless of the business sector.

Cash Conversion Cycle (CCC)

Cash Conversion Cycle (or Days of Net Working Capital) represents the time it takes to turn a dollar of investment in inventory into a dollar of cash from a sale. In other words, it is the period during which a company's capital is tied up and is unavailable for other uses.



Positive CCC

Indicates the company must finance its own operations, either through its own cash reserves or by borrowing money, while waiting for inventory to sell and customers to pay.

Negative CCC

Means the company collects money from customers before it has to pay its suppliers. In this scenario, the suppliers are effectively providing interest-free loans to fund the company's growth.

The Home Depot, Inc. Case (Efficiency Ratios)

6.Total Asset Turnover Ratio

- **$132,110/$60,908.5**
- **= 2.17**

- Average Total Assets
- =($70,581+$51,236)/2
- = $60,908.5

- For every dollar of assets Home Depot owned on average during 2020, they generated $2.17 in sales. Generally, a higher ratio indicates a more efficient use of the company's asset base.

7. DSO

- **($2,549/$132,110*)x365**
- **= 7 (rounded)**

- Average Net Receivables
- = ($2,992+$2,106)/2
- = $2,549

- A DSO of almost 7 days is exceptionally low for a major corporation and reflects Home Depot's business model as a retailer: the vast majority of their Net Sales are collected immediately at the register via cash or credit card, rather than through long-term invoicing.

** Net Sales (FY 2020) used as a proxy for Net Credit Sales*

8. DOI

- **($15,579/$87,257)x365**
- **= 65 (rounded)**

- Average Inventory
- = ($16,627+$14,531)/2
- = $15,579

- It takes Home Depot approximately 65 days to turn its inventory into a sale. A lower number generally suggests more efficient inventory management and less capital tied up in products sitting on shelves.

9. DPO

- **($9,696.5/$87,257)x365**
- **= 41 (rounded)**

- Average Accounts Payable
- = ($11,606+$7,787)/2
- = $9,696.5

- Home Depot's DPO of nearly 41 days highlights their significant bargaining power; by taking over a month to pay suppliers, they are essentially using “vendor financing” as an interest-free way to fund their growth.

On average, it takes Home Depot about (65+7-40=) 32 days from the moment they pay for their merchandise to the moment they receive the cash from the final sale. For a giant retailer, this is a very lean and efficient cycle.

5D. Liquidity ratios

Liquidity Ratios

10. Current Ratio

$$\frac{\text{Current Assets}}{\text{Current Liabilities}}$$

- Measures the extent to which a company's total current assets cover its total current liabilities on a specific date.
- A ratio < 1 generally raises concerns.
- The optimal level depends on the business environment in which a company operates.
- Broadly speaking, it should be between 1 and 2 (the time required to convert inventories and receivables into cash should be considered, as well as payments of taxes, dividends, etc.).
- What matters most is that it remains stable over the years.

11. Quick Ratio (Acid Test)

$$\frac{\text{Cash \& Cash Equivalents + Net Accounts Receivable + Marketable Securities}}{\text{Current Liabilities}}$$

- The current ratio is criticized because inventory is not readily convertible into cash.
- This ratio provides a more stringent test of short-term liquidity.
- It should not fall below 1, for the same reasons as the current ratio.

12. Cash Ratio

$$\frac{\text{Cash \& Cash Equivalents}}{\text{Current Liabilities}}$$

- It is the most stringent test of liquidity; it relates cash on hand to total current liabilities reported in the financial statements.
- It measures how much a company can pay off its current liabilities using only cash and cash equivalents.
- A cash ratio > 1 is unusual.

Liquidity Ratios

Liquidity ratios focus only on current assets (or specific components of current assets) and current liabilities.

These ratios are usually analyzed over time and/or compared with those of other companies operating in the same sector. Stability over the years and similarity across companies are generally appreciated by analysts.

Conversely, a changing ratio (either decreasing or increasing), or a ratio that differs from that of competitors, may serve as a warning signal. Examples:

1) *Company ABC's current ratio is higher than that of its competitors, while its quick ratio is the same. This suggests that inventory management is inefficient → attention should be paid to DOI.*
2) *Company ABC's cash ratio is decreasing. Is the company improving its DPO, or is it less able to meet its short-term obligations due to lower cash generation?*
3) *Company ABC's quick ratio is increasing. Is there any uncollectible receivable that has not been written off?*

The Home Depot, Inc. Case (Liquidity Ratios)

10. Current Ratio

- **$28,477/$22,709**
- **= 1.25**

- Home Depot has $1.25 in current assets for every $1 of current debt.

11. Quick Ratio

- **$11,850/$22,709**
- **= 0.52**

- Current Assets - Inventory
- =$28,477 - $16,627
- =$11,850

- Excluding inventory, the company has $0.52 of highly liquid assets per $1 of debt.

12. Cash Ratio

- **$7,895/$22,709**
- **= 0.35**

- 35% of current liabilities could be covered instantly with cash on hand.

5E. Solvency ratios

Solvency Ratios

Solvency ratios are very important for banks and external analysts because they represent, in a nutshell, the balance between liabilities and shareholders' equity. They also serve as a useful proxy for a company's ability to take on additional debt, especially in the case of the D/E ratio.

Most analysts focus only on interest-bearing debt when calculating solvency ratios. In this case, they compute ratios using the Net Financial Position (see slide set 1).

NFP = total Financial Liabilities – Cash and Cash equivalents

There is no optimal debt-to-equity structure applicable to all companies and/or sectors. However, broadly speaking, the higher the leverage, the greater the risk for the company.

Solvency Ratios

13. Debt-to-Equity Ratio (D/E)

$$\frac{\text{Total Liabilities}}{\text{Total Stockholders' Equity}}$$

- Expresses a company's total liabilities as a proportion of its stockholders' equity.
- A high ratio indicates that a company relies heavily on liabilities relative to equity financing. Heavy reliance on debt financing increases the risk that a company may not be able to meet its contractual financial obligations.
- This ratio is also influenced by a company's dividend payment policies.
- In this form, "total liabilities" include also non-interest-bearing debt, so it is influenced by working capital management.
- Broadly speaking, a D/E ratio between 2 and 4 is considered reasonable.

14. NFP-to-Equity Ratio (NFP/E)

$$\frac{\text{Net Financial Position}}{\text{Total Stockholders' Equity}}$$

- Shows the extent to which a company's net debt is supported by shareholders' equity.
- Focuses only on interest-bearing debt, as NFP excludes non-financial liabilities such as trade payables and accruals.
- Compared with the D/E ratio, NFP/E provides a clearer view of financial leverage because it isolates only financing-related obligations.
- A high ratio indicates greater reliance on debt financing and higher financial risk.
- A low ratio suggests a stronger capital structure and lower solvency risk.

Solvency Ratios

15. NFP-to-EBITDA Ratio

$$\frac{\text{Net Financial Position}}{\text{EBITDA}}$$

- It is one of the most widely used leverage metrics in debt covenant analysis and by lenders, investors, and rating agencies to assess financial sustainability.
- The ratio measures how many years of EBITDA would be required to repay the net financial position, assuming EBITDA is available for debt servicing (EBITDA as a proxy for operating cash flow).
- A lower ratio indicates stronger repayment capacity and lower financial risk, while a higher ratio reflects greater leverage and reliance on future operating performance.

16. Debt Service Coverage Ratio (DSCR)

$$\frac{\text{Operating Cash Flow}}{\text{Debt Service Cash Flow}}$$

- Debt Service Cash Flow consists of (interest paid) plus (principal repaid) during the year and represents the total cash outflows required for debt servicing.
- The ratio measures the company's forward-looking ability to repay its debt and, consequently, highlights the sustainability of its debt.
- Its value should be greater than 1 as the operating cash flows generated by the company should be sufficient at least to repay the financial liabilities and the related interest payable.
- In banking practice, the ratio is typically calculated as the ratio of EBITDA (as a proxy for operating cash flows) to total financial outflows for debt repayment (principal plus interest).

Why is EBITDA often considered a proxy for operating cash flows?

Because it represents the difference between operating revenues and operating expenses within the same accounting period, or in closely related periods in which such revenues are collected and costs are paid. Consequently, over a sufficiently long-time horizon, the average **EBITDA is regarded as broadly comparable to the average operating cash flow**.
However, EBITDA has the advantage of being a metric that is consistently and **readily calculable from financial statements**. In contrast, operating cash flow must be explicitly derived and may not always be directly available, particularly for companies that are not subject to mandatory cash flow statement reporting requirements.

The Home Depot, Inc. Case (Solvency Ratios)

13. D/E

- **$67,282/$3,299**
- **= 20.39**

- A ratio of 20.39 indicates a highly leveraged capital structure where the company relies significantly more on debt than equity to finance its assets.
- This exceptionally high figure is primarily driven by a small equity base, a result of Home Depot's aggressive historical share buyback programs, rather than financial distress. It reflects a deliberate strategy to optimize the cost of capital by using stable cash flows to support a high debt load.

14. NFP/E

- **$35,527/$3,299**
- **=10.77**

- NFP = 1,416 + 35,822 + 828 + 5,356 – 7,895 = $35,527
- A ratio of 10.77 means that for every dollar of equity Home Depot has, it carries $10.77 in net debt.
- This is considered very high for most industries, but it is a deliberate strategy for Home Depot. They use their massive cash flow to buy back shares (which lowers equity) and use cheap debt to fund operations, essentially betting on their own high efficiency to manage the interest payments.

15. NFP-to-EBITDA

- **$35,527/$20,406**
- **= 1.74**

- EBITDA = 44,853-24,447
- = $20,406
- A ratio of 1.74 is considered very healthy. It suggests that Home Depot could theoretically pay off its entire net debt in less than one and a half years using only its annual EBITDA.
- In the eyes of lenders and credit agencies, a ratio below 2.0 or 3.0 typically indicates strong solvency and a low risk of default, even for a highly leveraged company

16. DSCR

- **$18,628*/$1,805****
- **=10.32**

- A ratio of 10.32 indicates that Home Depot generates more than ten times the operating cash flow required to meet its annual debt service obligations.
- This reflects an exceptionally strong margin of safety and a very low risk of default, demonstrating that the company can easily cover its interest and principal payments while maintaining significant excess cash for reinvestment.

** Without a cash flow statement, to estimate operating cash flow, we use the average EBITDA over the three available years:*
[(44,853-24,447)+(37,572-19,740)+(37,160-19,513)]/3 =$18,628

***Without a cash flow statement, the Interest Expense is used as a direct proxy for interest paid. To estimate the principal paid during a year we look at the short-term debt, the current installments of long-term debt and the current portion of operating leases from the previous year's balance sheet. These figures represent the amount of debt that was legally due to be paid off within the following twelve months.*
Debt Service Cash Flow =1,347+974+1,839+828 =$1,805

Thank you!

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See you next time: FSA Part 3